condition. When access is granted, the organization must be able to demonstrate why the decision was permitted and identify the source of intent behind it. These qualities are essential for security reviews, audits, incident response, and regulatory compliance.

AI systems do not provide these guarantees. Given the same input, they can produce different outputs. Their explanations might be fluent but incomplete or even wrong. They infer intent instead of enforcing it. These traits are strengths in many situations, but they are liabilities in authorization.

At ACME, this distinction became clear during a design review. An engineer asked an AI tool to explain why a particular document access request was allowed. The explanation sounded correct and matched the policy engine's decision. However, when the team compared it to the actual policy evaluation trace, they discovered that the AI had not described the evaluation at all. The AI had inferred intent from the policy structure instead of analyzing the actual authorization logic. The mistake was caught during the review, but it revealed a deeper problem. Plausible explanations are not the same as correct decisions, and authorization can’t rely on interpretation alone.

That experience clarified an architectural boundary that ACME now treats as non-negotiable. Authorization decisions and their authoritative explanations come from the policy engine, which evaluates policies as code and data against clearly defined inputs and current system state. Given the same request and context, it produces the same result every time and can show exactly which rules were applied. AI systems operate differently, excelling at inference, summarization, and exploration. At ACME, AI is allowed to assist humans in understanding policies and decisions, but it is never allowed to substitute for policy evaluation or serve as the source of truth for why access was granted or denied.

Figure 17.2 shows the conceptual architecture ACME ultimately adopted to keep AI out of the authorization decision loop. In this architecture, authorization is done first, using policy and the current system state to determine what data and actions are permitted. That decision produces an authorized context, which is then used to construct prompts for the language model. The AI system never evaluates access rules and never directly queries